

Consider Natural number set N , every member of this set can be mapped into a representation in generic numeric base b .

$$\forall a \in N \rightarrow A(b), b \in N$$

And

$$A(b) = c_n b^n + c_{n-1} b^{n-1} + \dots + c_0 b^0 = a$$

Therefore, the c coefficient's, which belong to Natural set as well, can be calculated through:

$$c_n = \frac{a - c_{n+1} b^{n+1}}{b^n}$$

And, considering another number $c \in N$

$$C(b) = d_k b^k + d_{k-1} b^{k-1} + \dots + d_0 b^0 = c$$

We can then define the operator sum (+) between two numbers at the same base b

$$A(b) + C(b) = e_i b^i + e_{i-1} b^{i-1} + \dots + e_0 b^0 = e$$

$$e_0 = (c_0 + d_0) \text{ MOD } b$$

$$e_1 = ((c_1 + d_1) + \frac{(c_0 + d_0)}{b}) \text{ MOD } b$$

$$e_2 = ((c_2 + d_2) + \frac{((c_1 + d_1) + \frac{(c_0 + d_0)}{b})}{b}) \text{ MOD } b$$

...

$$e_n = (\sum_{i=0}^n \frac{c_{n-i} + d_{n-i}}{b^i}) \text{ MOD } b$$

A pseudocode can be easily written also, as shown next:

```
for(Carry=0 ; i<n ; i++)
```

```
{
```

```
    Ei = (Ci + Di + Carry)
```

```
    Carry = Ei/b
```

```
    Ei = Ei%b
```

```
}
```

Next, we can define the subtraction operation (-), when $a \geq c$, and both numbers are at the same base b , as seen below:

$$A(b) - C(b) = e_i b^i + e_{i-1} b^{i-1} + \dots + e_0 b^0 = e$$

$$e_0 = (b + c_0 - d_0) \text{ MOD } b$$

$$\bar{e}_n = (b + c_n - d_n)$$

Note that, if the subtraction has negative carry, $\bar{e}_n < b$, otherwise, $\bar{e}_n \geq b$

$$e_1 = \left(b + c_1 - d_1 - \left(1 - \frac{(b + c_0 - d_0)}{b} \right) \right) \text{ MOD } b$$

The Negative carry is represented by the expression bellow

$$K_1 = \frac{(b + c_0 - d_0)}{b} = \frac{\bar{e}_0}{b}$$

It also can be written as

$$e_1 = \left(\bar{e}_1 - \left(1 - \frac{\bar{e}_0}{b} \right) \right) \text{ MOD } b$$

$$e_2 = \left(\bar{e}_2 - \left(1 - \frac{\left(\bar{e}_2 - \left(1 - \frac{\bar{e}_0}{b} \right) \right)}{b} \right) \right) \text{ MOD } b$$

$$K_2 = \frac{\left(\bar{e}_2 - \left(1 - \frac{\bar{e}_0}{b} \right) \right)}{b}$$

$$e_3 = \left(\bar{e}_3 - \left(1 - \frac{\bar{e}_2 - \left(1 - \frac{\left(\bar{e}_1 - \left(1 - \frac{\bar{e}_0}{b} \right) \right)}{b} \right)}{b} \right) \right) \text{ MOD } b$$

$$K_3 = \frac{\bar{e}_2 - \left(1 - \frac{\left(\bar{e}_1 - \left(1 - \frac{\bar{e}_0}{b} \right) \right)}{b} \right)}{b}$$

$$e_n = \left(\bar{e}_n - (1 - K_n) \right) \text{ MOD } b$$

A pseudocode can be easily written also, as shown next:

```
for(Carry=1 ; i<n ; i++)
{
    Ei = (b+Ci - Di-(1-Carry))
    Carry = Ei/b
    Ei = Ei%b
}
```

With that in mind, defining multiplication operation (*), between two number at the same base b, as seen below:

$$A(b) * C(b) = e_i b^i + e_{i-1} b^{i-1} + \dots + e_0 b^0 = e$$

We can define the first digit e_0 :

$$e_0 = (c_0 * d_0) \text{ MOD } b$$

And also, it's carry $\bar{e}_0$:

$$\bar{e}_0 = \frac{(c_0 * d_0)}{b}$$

Then, e_1 can be defined as:

$$e_1 = (c_0 * d_1 + c_1 * d_0 + \bar{e}_0) \text{ MOD } b$$

Also, $\bar{e}_1$ can be defined as:

$$\bar{e}_1 = \frac{(c_0 * d_1 + c_1 * d_0 + \bar{e}_0)}{b}$$

And then, further extending it for next coefficients:

$$e_2 = (c_0 * d_2 + c_1 * d_1 + c_2 * d_0 + \bar{e}_1) \text{ MOD } b$$

$$\bar{e}_2 = \frac{(c_0 * d_2 + c_1 * d_1 + c_2 * d_0 + \bar{e}_1)}{b}$$

$$e_3 = (c_0 * d_3 + c_1 * d_2 + c_2 * d_1 + c_3 * d_0 + \bar{e}_2) \text{ MOD } b$$

$$\bar{e}_3 = \frac{(c_0 * d_3 + c_1 * d_2 + c_2 * d_1 + c_3 * d_0 + \bar{e}_2)}{b}$$

...

$$e_n = (c_0 * d_n + c_1 * d_{n-1} + \dots + c_n * d_0 + \bar{e}_{n-1}) \text{ MOD } b$$

$$\bar{e}_n = \frac{(c_0 * d_n + c_1 * d_{n-1} + \dots + c_n * d_0 + \bar{e}_{n-1})}{b}$$

So generically, every single digit can be defined as follows:

$$e_n = \left(c_0 * d_n + c_1 * d_{n-1} + \cdots + c_n * d_0 + \frac{(c_0 * d_{n-1} + c_1 * d_{n-2} + \cdots + c_{n-1} * d_0 + \bar{e}_{n-1})}{b} \right) MOD \, b$$